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1) Resources

The agent has to run on an Ubuntu 24.04 system. You need to document any requirements for compiling and running your code on a fresh installation.

To run the code, I used both:

- Clingo located in a venv using Miniconda Terminal (using input.lp files with input grid)
- ASPChef website
- the ASP extension for Visual Studio code (using an hardcoded input inside the code)

I used on my device the clingo version 5.7.1

2) Problem statement

The main task of the project is:

Implement an automated Cross-out puzzle solver using at least one of the formalisms covered in this module.

A cross-out puzzle consists of a **rectangular grid of symbols**.

For a solution, some symbols should be crossed (cancellati) out such that all of the following conditions are met:

1. No symbol should appear more than once in a row or a column (crossed-out occurrences do not count)
2. Crossed-out symbols must not occur next to each other horizontally or vertically.
3. All symbols that are not crossed out should form a contiguous area (considering horizontal and vertical connections). That means that each non-crossed-out symbol must be reached from all other non-crossed-out symbols by moving only horizontally or vertically over non-crossed-out symbols.

3) Code and solution

I decided to use the Answer Set Programming formalism, code can be found in project.lp.

To run the code, use this command:

```
clingo project.lp input.lp
```

Doing so, you will obtain models in output.

In the context of the given input for this project (so input.lp file), we can find only 1 model.

3.a) choosing the method(s)

To prohibit a symbol from appearing more than once in the same row or column of the grid, I have chosen to use a choice rule so that each cell is assigned to one of two sets.

These are the cells I want to keep, and those I want to tick with a cross.

In doing so, if it is possible to identify two cells that contain the same symbol and which you have chosen to keep, then it must be forbidden for them to be in the same row or column.

To ensure that crossed-out symbols must not occur next to each other, it was necessary to define the notion of adjacency between two cells.

Note that this must be made symmetrical.

Having done this, if it is possible to identify two cells that you have chosen as crossedout, then they cannot be adjacent.

Finally, we want to ensure that cells that the solver decided to keep, form a contiguous area.

The idea is to define a predicate that signals reachability between two cells, so to ensure that it is possible to reach one cell from the other. This must be done on both the vertical and horizontal axis.

Then if for a cell the crossedout predicate is false (and therefore I have chosen to keep that cell), then it is mandatory that the cell is reachable from at least one other cell that I have chosen to keep.

I finally define a last predicate `is_valid_keep_cell`, it is true if a cell for which keep is true (i.e. I have chosen to keep that cell) has at least one other adjacent cell for which in turn the keep predicate is true.

Finally I forbid the case where, for a cell, the predicate keep is true but `is_valid_keep_cell` is not.

3.b) testing

I have chosen to create several examples of input that can be used to evaluate the correct functioning of the programme.

Try first the original input given for this project running :

```
clingo project.lp input.lp 0
```

Then try to use these other input files that use a not square matrix

```
clingo project.lp input_1.lp 0
```

```
clingo project.lp input_2.lp 0
```

4) Output

Here to demonstrate, I used the original project input given (input.lp file).

Output using **VScode extension** :

```

PS D:\Specialistica\Intro to AI Part A\Progetto esame> c:\Users\User\.vscode\extensions\ffrankreiter.answer-set-programming-language-support-0.7.1\clingo_win.exe "d:\Specialistica\Intro to AI Part A\Progetto esame\project_edoardo_diana.lp" 0
5917fclingo version 5.4.0
Reading from ...rogetto esame\project_edoardo_diana.lp
Solving...
Answer: 1
crossedout(1,3,b) crossedout(1,5,c) crossedout(2,2,e) crossedout(3,3,a) crossedout(3,5,e) crossedout(4,1,c) crossedout(5,3,a) crossedout(5,5,a)
SATISFIABLE

Models      : 1
Calls       : 1
Time        : 0.008s (Solving: 0.00s 1st Model: 0.00s Unsat: 0.00s)
CPU Time    : 0.000s

```

Output using **ASPChef** :

Height	200	Decode	__base64__	ECHO	USE CONSTRAINTS	Input	models																									
<pre> X2, Y2). 40 :- not is_valid_keeped_cell(X, Y, S), keep(X, Y, S). 41 42 %%% 43 %%% 44 #show crossedout/3. 45 #show cell/3. 46 </pre>						<pre> 1 c e b b c 2 b e a d e 3 a d a c e 4 c a b e c 5 e b a a a </pre>																										
# of models 0						RAISE ERROR																										
model 1 of 1																																
Height	150	Predicate	__graph__	ECHO		Ready! <pre> 18 cell(4,3,"b"). 19 cell(4,4,"e"). 20 cell(4,5,"c"). 21 cell(5,1,"e"). 22 cell(5,2,"b"). 23 cell(5,3,"a"). 24 cell(5,4,"a"). 25 cell(5,5,"a"). 26 crossedout(1,3,"b"). 27 crossedout(1,5,"c"). 28 crossedout(2,2,"e"). 29 crossedout(3,3,"a"). 30 crossedout(3,5,"e"). 31 crossedout(4,1,"c"). 32 crossedout(5,3,"a"). 33 crossedout(5,5,"a"). </pre>																										
Highlight color	yellow		Highlight text color	red																												
<table border="1"> <tr><td>c</td><td>e</td><td>b</td><td>b</td><td>c</td></tr> <tr><td>b</td><td>e</td><td>a</td><td>d</td><td>e</td></tr> <tr><td>a</td><td>d</td><td>a</td><td>c</td><td>e</td></tr> <tr><td>c</td><td>a</td><td>b</td><td>e</td><td>c</td></tr> <tr><td>e</td><td>b</td><td>a</td><td>a</td><td>a</td></tr> </table>								c	e	b	b	c	b	e	a	d	e	a	d	a	c	e	c	a	b	e	c	e	b	a	a	a
c	e	b	b	c																												
b	e	a	d	e																												
a	d	a	c	e																												
c	a	b	e	c																												
e	b	a	a	a																												
Search	Search...																															

Output using **miniconda**

```

(base) D:\Specialistica\Intro to AI Part A\Progetto esame>conda activate potassco
(potassco) D:\Specialistica\Intro to AI Part A\Progetto esame>clingo project_edoardo_diana.lp input.lp 0
clingo version 5.7.1
Reading from project_edoardo_diana.lp ...
Solving...
Answer: 1
crossedout(1,3,b) crossedout(1,5,c) crossedout(2,2,e) crossedout(3,3,a) crossedout(3,5,e) crossedout(4,1,c) crossedout(5,3,a) crossedout(5,5,a)
SATISFIABLE

Models      : 1
Calls       : 1
Time        : 0.029s (Solving: 0.00s 1st Model: 0.00s Unsat: 0.00s)
CPU Time    : 0.000s

```

5) Considerations about Ethics

If a variant of my solver were implemented outside of this project/game, there would be some practices that could benefit from this.

5.a) Steganography

In the military sphere, an example of application could be related to steganography.

Unlike cryptography as we understand it today, steganography is a technique that aims to conceal communication between two interlocutors.

It would therefore be possible to implement a system to communicate publicly, without worrying that an adversary, while obtaining the message, might learn its true meaning.

W	R	E	H	E
W	R	E	E	E
R	U	A	A	A
R	E	E	A	E
W	R	R	H	R

W	R	E	H	E
W	R	E	E	E
R	U	A	A	A
R	E	E	A	E
W	R	R	H	R

Since he does not know the methodology for eliminating superfluous symbols, it is very difficult for him to find out.

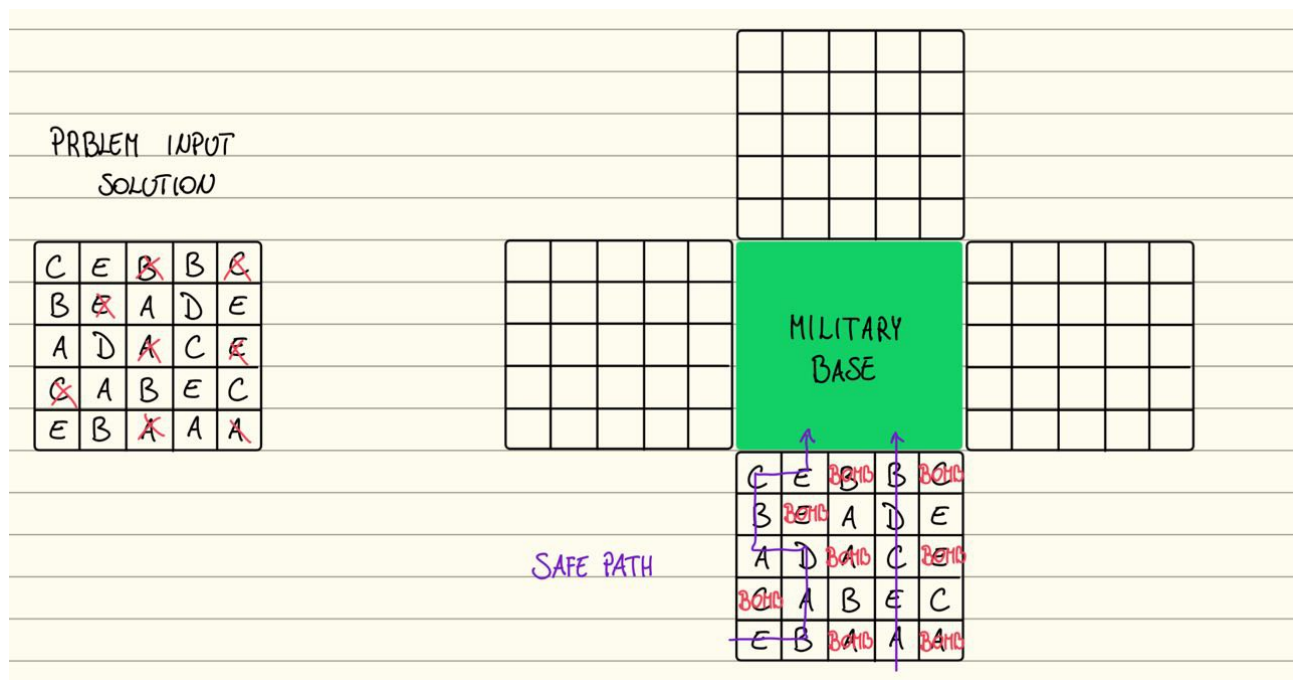
5.b) Placement of mines

Another idea could be related to the arrangement of mines around a military base.

If we imagine this as a square, we can also consider the 4 areas adjacent to it (vertically and horizontally) as squares.

For each of these 4 areas, it is possible to obtain an arrangement that hides a safe passage to the base.

Allied soldiers will in fact be able to cross the minefield easily, since they are aware of the methodology to be followed (i.e. the 3 requirements to be met in this project).



3.c) Ethical and rational thinking

I have analysed two possible scenarios that could be useful about the exchange of information, which is crucial in a military context.

However, there are certainly other possible applications that I have not thought of, but which could act more decisively on certain occasions.

When I think of the military, the first thought that comes to my mind is related to armaments, to the lives at stake on both sides, but especially to the lives of those who are totally unrelated to this.

Glaring examples of this are the recent conflicts, where the main ones to pay the price are ordinary citizens.

The constant arms race is now totally out of control. To see nations spending more money on armaments than on their own nation's health is something surreal.

And in this race, the latest hot topic is obviously artificial intelligence. By dint of wanting to implement it just about everywhere, there is certainly a risk of getting the wrong end of the stick.

A tool designed by the programmer for one purpose, may be revolutionised from another perspective, thus becoming a means of panic and despair in the hands of someone not very conscious of using it.

In the face of unexpected consequences, who will take the blame?

Is the process of technological advancement ethically wrong, or should those who decide to wield this as a weapon do so with things like fundamental human rights in mind?

Self-guided drones, tracking missiles, cybersecurity attacks. Artificial intelligence may be applicable to all this, but when the damage is done who will take responsibility for the consequences?

In my opinion, it is imperative and compulsory to have self-critical thinking, a thinking brain capable of truly assessing the implications of one's decisions. Reasoning from an ethical point of view could instil doubts about one's actions, allowing for a more sympathetic reappraisal.

Edoardo Diana